

MODEL PAPER

Sreenidhi Institute of Science and Technology

(An Autonomous Institution)

Regulations: A22

Code No: 9EC05

Date:

B.Tech III-Year I-Semester External Examination, (Regular)
(Computer Networks)

Time: 3 Hours

Max.Marks:60

Note: a) No additional answer sheets will be provided.
b) All sub-parts of a question must be answered at one place only, otherwise it will not be valued.
c) Missing data can be assumed suitably.

Bloom's Cognitive Levels of Learning (BCLL)

Remember	L1	Apply	L3	Evaluate	L5
Understand	L2	Analyze	L4	Create	L6

Part - A

6x2=12 Marks

ANSWER ALL QUESTIONS

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-I	Define topology and mention the types of topologies.	L1	CO1	[2M]
2	From Unit-II	What is Piggy backing? Illustrate with a neat diagram?	L2	CO2	[2M]
3	From Unit-III	What are the uses of bridges in networks?	L1	CO3	[2M]
4	From Unit-IV	List the different dynamic routing algorithms.	L1	CO4	[2M]
5	From Unit-V	What is congestion?	L1	CO4	[2M]
6	From Unit-VI	What is DNS? Give example.	L2	CO5	[2M]

Part – B

6x8=48 Marks

ANSWER ALL QUESTIONS.

S.No		Coverage		BCLL	CO(s)	Marks
7.		From Unit-I	Explain OSI Reference Model in detail.	L2	CO1	[8M]
			OR			
8.		From Unit-I	What is Guided media? Explain its types with neat diagrams.	L2	CO1	[8M]
9.		From Unit-II	Define Checksum. Calculate the checksum for given $M(x) = 1101011011$ and the generator polynomial is $g(x)=x^4+x+1$ using CRC at the sending side, verify the checksum at receiving side	L6	CO2	[8M]
			OR			
10.		From Unit-II	Analyze the behavior of the Go-Back-N Sliding Window Protocol when a single frame is lost. Use a neat diagram to illustrate how retransmissions occur and compare it with Selective Repeat.	L4	CO2	[8M]
11.		From Unit-III	What is Pure ALOHA and slotted ALOHA? Consider the delay of both at low load which one is less? Explain your answer.	L4	CO3	[8M]
			OR			
12.		From Unit-III	Demonstrate CSMA protocols with neat sketch.	L3	CO3	[8M]
13.		From Unit-IV	What is Distance Vector Routing Algorithm? Why count to infinity problem occurs in this algorithm.	L2	CO4	[8M]
			OR			
14.		From Unit-IV	State and explain briefly about a Dijkstra algorithm illustrating with an example.	L2	CO4	[8M]
15.		From Unit-V	What is IPv4? Draw IPV4 header and explain each field.	L2	CO4	[8M]
			OR			

16.		From Unit-V	How Leakey bucket and token bucket relates to Quality of Service?	L4	CO4	[8M]
17.		From Unit-VI	Illustrate TCP connection establishment using Three way Handshake.	L2	CO5	[8M]
			OR			
18.		From Unit-VI	Describe how FTP is used to upload and download files in a real-world scenario such as a web server..	L3	CO5	[8M]

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Note: In Part - B Questions under a) OR b) can be further divided into sub questions if required.